

Test Cases & Test Report

Credit Card Approval Prediction System | Project Testing Phase

Testing Scope

Testing covers: form input validation, Pipeline prediction for both outcomes, friendly-to-raw days conversion, error handling, result page rendering, and the toggle mode switch.

Test Cases

ID	Description	Input Summary	Expected Result
TC-01	Approved applicant — high income, stable employment, own car+realty, result=150000, High Credit Card Approval	Female, 45yrs, employed, own car+realty, income=150000, High Credit Card Approval	Approved
TC-02	Rejected applicant — poor credit history, single, no car, income=30000, Secondary education, STATUS=Rejected	Male, 45yrs, single, no car, income=30000, Secondary education, STATUS=Rejected	Rejected
TC-03	Friendly age input conversion	Age=35 entered in friendly mode	JS converts to DAYS_BIRTH = -Math.round(35 × 365.25)
TC-04	Unemployed/retired experience	Employment experience = 0 in friendly mode	DAYS_EMPLOYED set to 365243 (UNDP sentinel value)
TC-05	Raw days mode toggle	User clicks 'Switch to Raw Days Mode'	Friendly inputs hidden; raw DAYS_BIRTH/DAYS_EMPLOYED shown
TC-06	Missing required field	AMT_INCOME_TOTAL left blank	HTML5 form validation triggers; page does not submit
TC-07	Server-side exception	Invalid/corrupt form data bypasses client validation	result.html renders error card (■) with error message; 200 response
TC-08	Unknown occupation	OCCUPATION_TYPE not selected (or selected but blank)	OneHotEncoder handle_unknown='ignore' handles it
TC-09	Pipeline column order	Input DataFrame columns match FEATURE_COLUMNS order	Pipeline predicts correctly from no feature mismatch error
TC-10	Return to form	User clicks '← Analyze Another Applicant'	Redirected to /index.html; form is empty and ready

Test Summary Report

ID	Status	Notes
TC-01	Pass	Green tick result card rendered with correct message.
TC-02	Pass	Red cross result card rendered with risk message.
TC-03	Pass	JS conversion verified: 35 years → -12784 days.
TC-04	Pass	365243 sentinel value correctly sent for 0 years experience.
TC-05	Pass	Toggle hides/shows correct input groups; required attributes swapped.
TC-06	Pass	HTML5 required attribute prevents submission without income.
TC-07	Pass	Exception caught; error card shown; 200 response from Flask.
TC-08	Pass	Unknown occupation handled silently by OHE; prediction succeeds.
TC-09	Pass	DataFrame built with FEATURE_COLUMNS order; pipeline accepts it.
TC-10	Pass	Back button navigates to / correctly.

Conclusion

All 10 test cases passed. The system correctly predicts credit approval/rejection, handles friendly and raw input modes, converts age/experience to negative-day format, and recovers gracefully from invalid input.